

Jiaru Li

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Research Interests

My research combines analytical theory and numerical simulations to study planetary dynamics, disk–planet interactions, hydrodynamics of protoplanetary disks, and gravitational-wave sources in AGN disks.

Education

Cornell University, Ithaca, NY, USA *Sept. 2017 - May 2023*
 - **Ph.D.** in Astronomy and Space Sciences, May 2023
 ◦ **Advisor:** Prof. Dong Lai
 ◦ **Thesis:** Dynamical Evolution of Multi-Orbiter Systems: Application to Planets, Protoplanetary Disks, and Black Holes
 - **M.S.** in Astronomy, Dec 2019

University of Toronto at Scarborough, Toronto, ON, Canada *Sept. 2013 - May 2017*
 - **Honours B.Sc.** in Physics (Specialist) and Mathematics (Major), May 2017

Experience

CIERA Postdoctoral Fellow *2023 - present*
Center for Interdisciplinary Exploration and Research in Astrophysics, Northwestern University
Independent Prized Fellow

Graduate Research Assistant *2017 - 2023*
Department of Astronomy and Space Sciences, Cornell University
Advisor: Prof. Dong Lai

CSES Graduate Student Fellow *2020 - 2022*
Theoretical Division, Los Alamos National Laboratory
Mentor: Dr. Hui Li

UTEA Undergraduate Research Assistant *2015 - 2017*
Department of Physical and Environmental Sciences, University of Toronto at Scarborough
Supervisor: Prof. Artur Izmaylov

Selected Honors and Awards

CIERA Postdoctoral Fellowship, Northwestern University *2023*
 Competitive postdoctoral fellowship supporting independent research at CIERA.

CSES Student Fellowship, Los Alamos National Laboratory (LANL) *2020*
 Graduate fellowship supporting research collaborations with scientists in LANL's Theoretical Division.

New Graduate Student Fellowship, Cornell University *2017*
 University fellowship providing one year of funding for outstanding incoming graduate students.

Governor General's Silver Medal Nomination, University of Toronto *2017*
 Nominated by the Scarborough campus as the graduating student with the highest academic standing.

Samuel Beatty In-Course Scholarship, University of Toronto *2017*
 Awarded to top students in Mathematics, Physics, Statistics, or Computer Science based on academic excellence.

University of Toronto Excellence Award *2016*
 Supports exceptional undergraduates to perform faculty-supervised research projects.

E-Fund Scholarship, University of Toronto at Scarborough 2013 - 2016
 Merit-based scholarship for Chinese students in the Green Path program with outstanding academic achievement.

Vincent Bladen Scholarship, University of Toronto at Scarborough 2015
 Awarded to two students per year for exceptional academic achievement.

A. D. Allen Memorial Scholarship, University of Toronto at Scarborough 2014
 Awarded to the most outstanding student of the year across all fields of study.

Selected Mentorship and Teaching Experience

Students Mentored:

Jupiter Ding, graduate student at Northwestern University 2025 - present
Jesse Richter, undergraduate student at Northwestern University 2024 - present
Dieran Wang, former undergraduate student at Shandong University, now Ph.D. student at Shanghai Jiao Tong University 2024 - present
Kecheng Qian, former undergraduate student at Cornell University, now Ph.D. student at University of California, Berkeley 2023 - 2024

Teaching Experience:

Teaching Assistant, Cornell University and University of Toronto 2016 - 2019
 Courses: *Our Solar System*, *From New Worlds to Black Holes*, *Calculus I & II*, *Physics I & II*, *Linear Algebra II*, and *Discrete Mathematics*

Selected Presentations

Invited Talks

- 01/2026 - **Colloquium**: Department of Astronomy Colloquium, Westlake University, Hangzhou, China
- 12/2025 - **Conference**: International Conference on Exoplanets and Planet Formation, Shanghai, China
- 10/2024 - **Seminar**: Theoretical Astrophysics Center Seminar, University of California, Berkeley
- 10/2024 - **Conference**: Transient Phenomena and Physical Processes Around Supermassive Black Holes, Tsung-Dao Lee Institute, China
- 09/2024 - **Seminar**: Department of Astronomy Tea Talk, Indiana University Bloomington
- 08/2024 - **Conference**: New Ideas on the Origin of Black Hole Mergers, Niels Bohr Institute, Denmark
- 03/2024 - **Seminar**: Center for Theory and Computation Seminar, University of Maryland
- 01/2024 - **Seminar**: ET Science Seminar, Shanghai Astronomical Observatory, Chinese Academy of Sciences, China
- 12/2023 - **Conference**: The 32nd Texas Symposium on Relativistic Astrophysics, Shanghai, China
- 03/2023 - **Conference**: AGN Santa Fe: "Where are the Objects in AGN Disks", Santa Fe, NM
- 11/2022 - **Seminar**: Center for Exoplanets and Habitable Worlds Seminar, Penn State University
- 10/2022 - **Seminar**: Center for Relativistic Astrophysics Seminar, Georgia Tech

Other Talks

- 05/2025 - **Conference**: The 56th AAS Division on Dynamical Astronomy Meeting, Georgia Tech
- 05/2024 - **Conference**: The 55th AAS Division on Dynamical Astronomy Meeting, University of Toronto
- 12/2023 - **Conference**: Exoplanets and Planet Formation Workshop, Beijing, China
- 10/2022 - **Workshop**: Recent Advances in Supermassive Black Holes, Cornell University
- 08/2022 - **Seminar**: Astrophysics Seminar, Los Alamos National Laboratory
- 05/2021 - **Conference**: Distorted Astrophysical Discs, Kavli Institute for Cosmology, Cambridge, UK

Selected Service

Reviewer (2023–present) — Astrophysical Journal and Letters, MNRAS, Astronomy & Astrophysics.
Panelist reviewer (2024) — NASA research program proposals.
SOC member (2025) — CIERA Fellows at 15 conference.

List of Publications

Overview:

- Total of **18 papers**: including **10 as first author**, plus **2 as co-first author**, and **2 led by students**.
- Astrophysics (published + to be submitted soon): 12+1; Computer Science: 2+1; Quantum Physics: 2.
- Citations: 400+ as of April 2026. Full details on [Google Scholar](#)  and [the Astrophysics Data System](#) .

First-Author and Mentored Papers (*indicates students supervised by J.L.)

- **Li** & Lithwick, *Steady Warps: Linear, Nonlinear, and Breaking*, to be submitted soon 
- **Li**, O'Connor, & Rasio, *Intruder Alert: Breaking resonant chains with planetesimal flybys*, 2026, ApJL, 998, L5
- Wang*, **Li**, & Lai, *Spin and Obliquity Distributions of Low-mass Planets Shaped by Dynamical Instability*, 2026, ApJ, 996, 3
- **Li**, Rodet, & Lai, *Dynamical instability in multi-orbiter systems with gas friction*, 2024, MNRAS, 528, 1198
- Qian*, **Li**, & Lai, *Dynamical Friction Models for Black Hole Binary Formation in Active Galactic Nucleus Disks*, 2024, ApJ, 962, 143
- **Li** & Lai, *Resonant Excitation of Planetary Eccentricity due to a Dispersing Eccentric Protoplanetary Disk: A New Mechanism of Generating Large Planetary Eccentricities*, 2023, ApJ, 956, 17
- **Li**, Dempsey, Li, Lai, & Li, *Hydrodynamical Simulations of Black Hole Binary Formation in AGN Disks*, 2023, ApJL, 944, L42
- **Li**, Lai, & Rodet, *Long-term Evolution of Tightly Packed Stellar Black Holes in AGN Disks: Formation of Merging Black Hole Binaries via Close Encounters*, 2022, ApJ, 934, 154
- **Li**, Dempsey, Li, & Li, *Ring Formation in Protoplanetary Disks Driven by an Eccentric Instability*, 2021, ApJ, 910, 79
- **Li**, Lai, Anderson, & Pu, *Giant planet scatterings and collisions: hydrodynamics, merger-ejection branching ratio, and properties of the remnants*, 2021, MNRAS, 501, 1621
- **Li** & Lai, *Planetary Spin and Obliquity from Mergers*, 2020, ApJL, 898, L20

Other Astrophysics Papers

- Li, Dempsey, Li, Li, & **Li**, *Hot Circumsingle Disks Drive Binary Black Hole Mergers in Active Galactic Nucleus Disks*, 2022, ApJL, 928, L19
- Li, Dempsey, Li, Li, & **Li**, *Orbital Evolution of Binary Black Holes in Active Galactic Nucleus Disks: A Disk Channel for Binary Black Hole Mergers?*, 2021, ApJ, 911, 124

Computer Science and Quantum Physics Papers

- Cao, **Li** (co-first author), Zhao, Zhang, Sun, Penn, *ConstellationBench: Evaluating Astronomical Reasoning and Spatial Intelligence in Vision-Language Models*, 2026, Conference on Language Modeling (COLM 2026, submitted)
- Lu, **Li** (co-first author), Zhao, Chen, Wang, Cai, Li, He, *On the Representation Geometry of LoRA Model Merging*, 2026, The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)
- Zhao, Zhang, **Li**, Niu, He, Min, Penn, *Tiny Budgets, Big Gains: Parameter Placement Strategy in Parameter Super-Efficient Fine-Tuning*, 2025, Empirical Methods in Natural Language Processing (EMNLP 2025)
- **Li**, Joubert-Doriol, & Izmaylov, *Geometric phase effects in excited state dynamics through a conical intersection in large molecules: N-dimensional linear vibronic coupling model study*, 2017, The Journal of Chemical Physics, 147, 064106
- Izmaylov, **Li**, & Joubert-Doriol, *Adiabatic Definition of Geometric Phase Effects*, 2016, Journal of Chemical Theory and Computation, 12, 5278-5283